

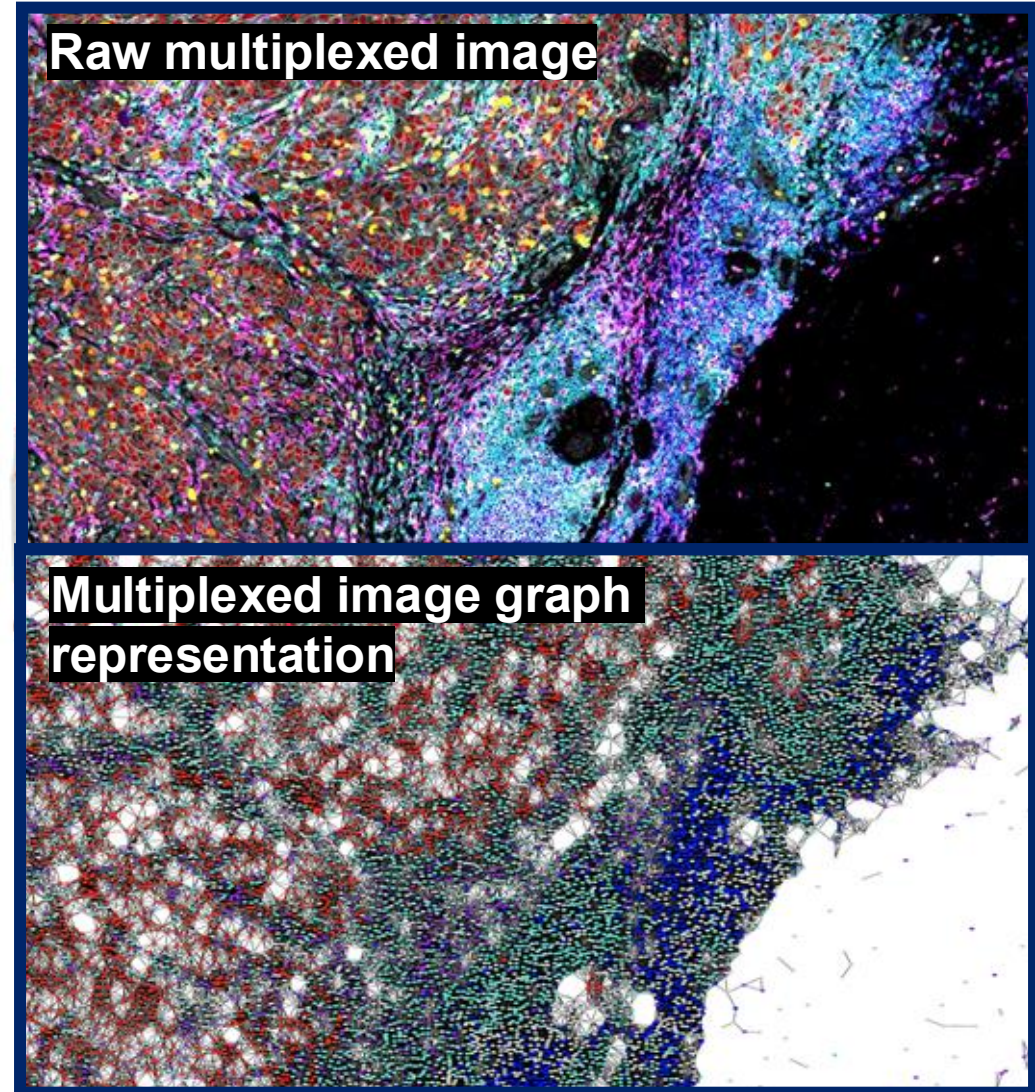
Cell graphs and neighborhood clustering

Day 6: Spatial data

Maryam Pourmaleki, PhD | Stanford University

Motivation

- Spatial omics data is highly complex and high-dimensional
- Supervised spatial analyses fail to capture the full breadth of data
- Graphs have been utilized to represent relationships for years (e.g., social network)
- Spatial data as graphs
- Nodes (cells) and edges (distances)
- **Graphs** enable a rich suite of topological measurements on complex spatial data
 - Supervised (e.g., degree)
 - Unsupervised (e.g., GNN)



What is a Graph?

A graph is a mathematical structure made of two things:

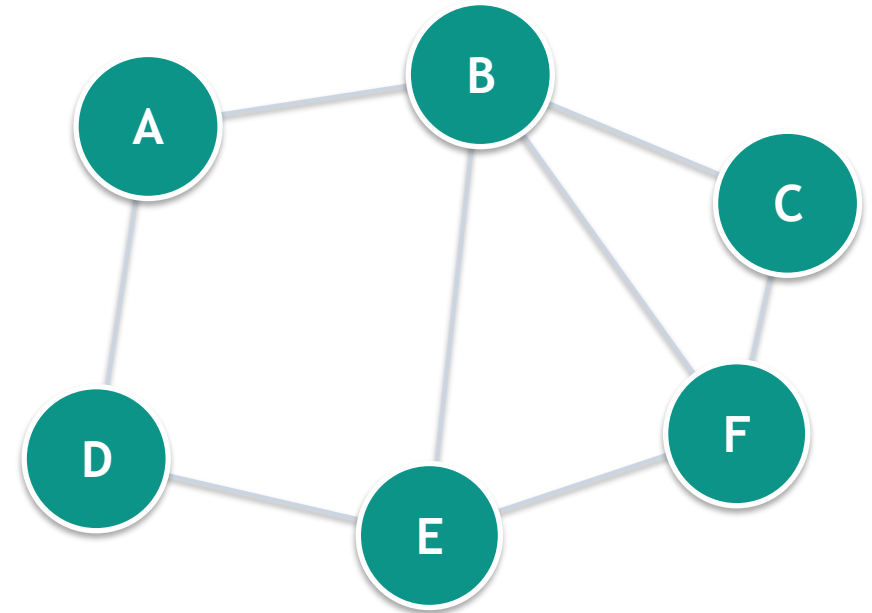
Nodes (Vertices)

The "things" in your network — objects, entities, data points

Edges (Links)

Connections between nodes — relationships, proximity, interactions

Together they capture both the identity of objects AND how they relate to each other.

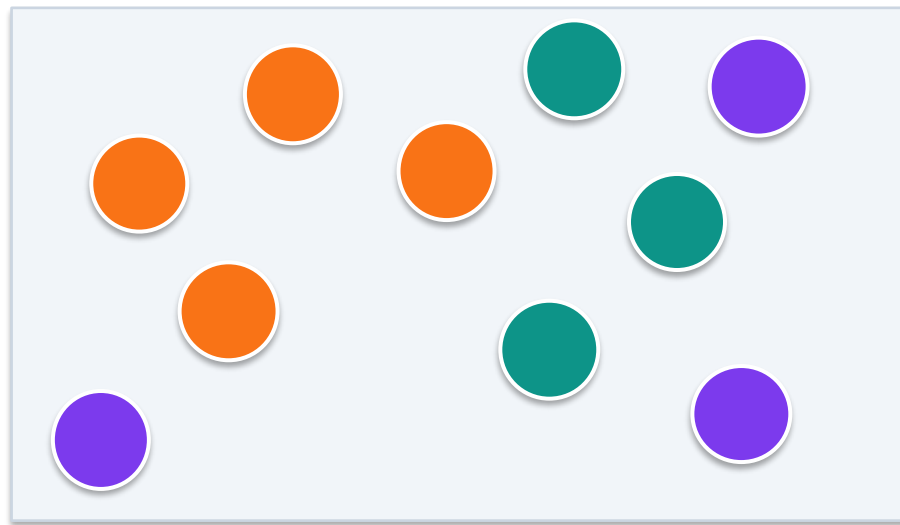


$$G = (V, E)$$

Vertices + Edges

Cells as Nodes, Connections as Edges

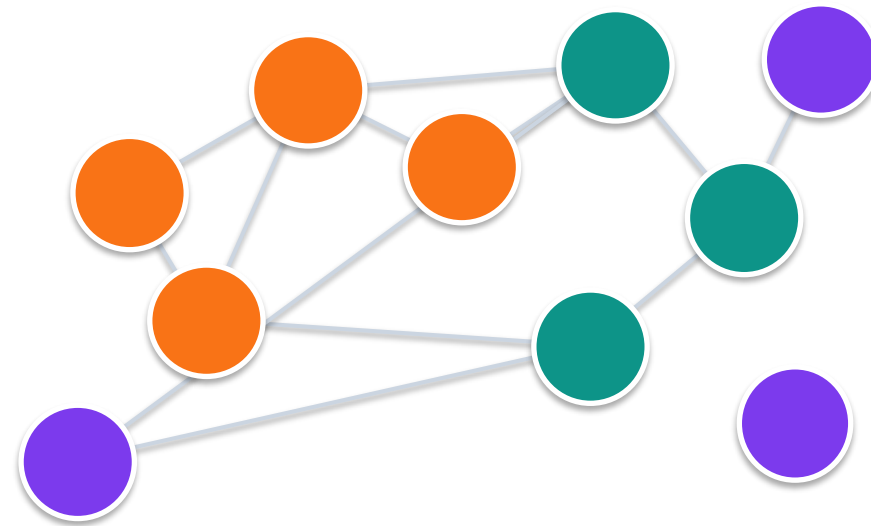
In spatial biology, every cell becomes a node — its gene expression profile carries its 'features'. Physical proximity defines the edges.



Tissue Section



build graph



Spatial Graph



Tumor cell



Immune cell



Stromal cell

Why Graphs for Spatial Data?



Captures Neighbourhood Structure

Encodes which cells are physically close to each other — information lost in expression matrices.



Models Cell-Cell Interactions

Edges represent signalling relationships, mechanical contact, or shared microenvironment.



Flexible & Platform-Agnostic

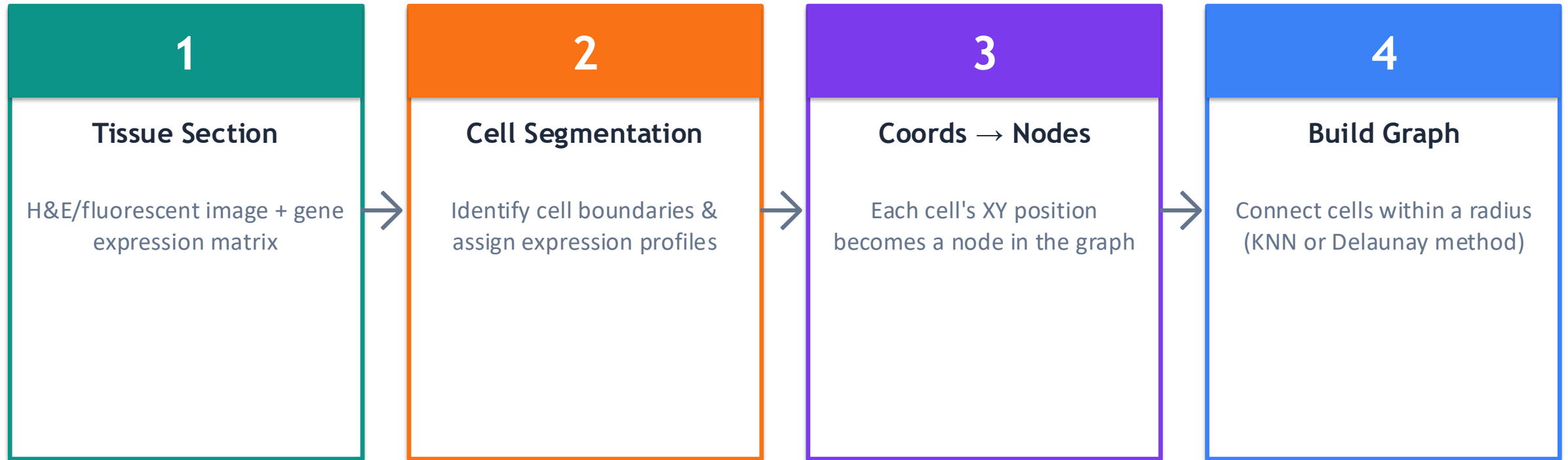
Works for Visium, Xenium, MERFISH, and more — no regular grid required.



Unlocks Powerful Algorithms

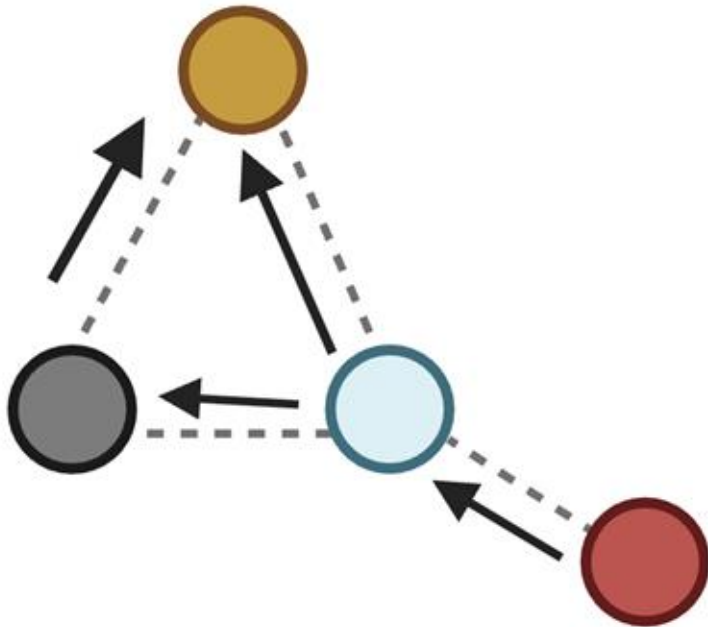
Graph metrics and GNNs reveal biology that is simply invisible to tabular methods.

From Tissue Section to Spatial Graph

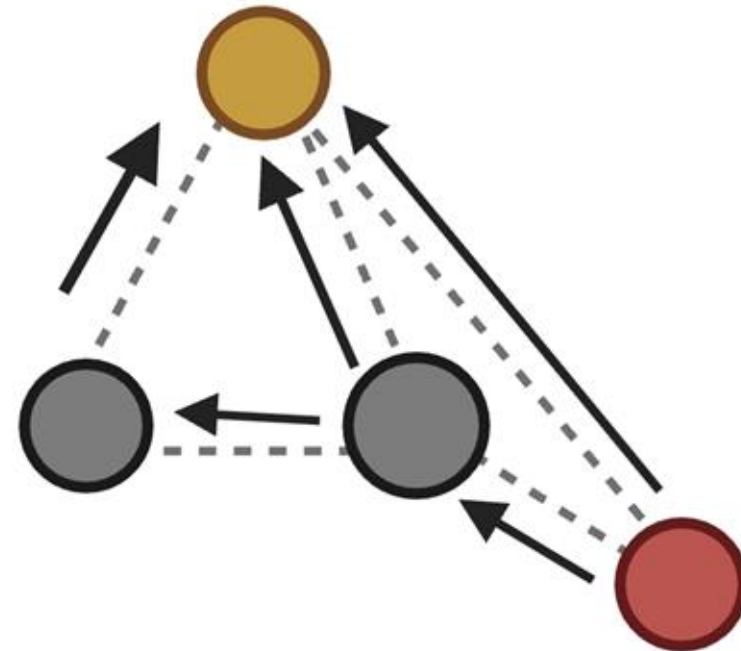


Considerations for building the graph: defining the adjacency matrix

Delaunay



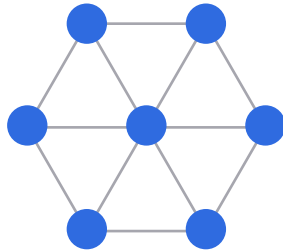
Distance



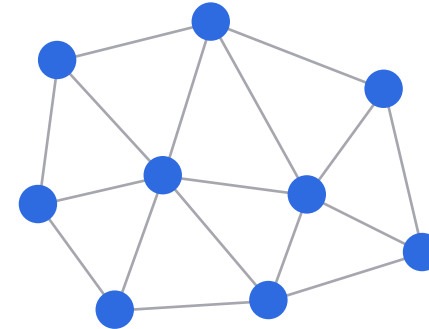
Other considerations: *Spots versus cells*

The graph is built differently because the geometry is different — a regular grid versus an irregular point cloud.

Visium — spots on a grid



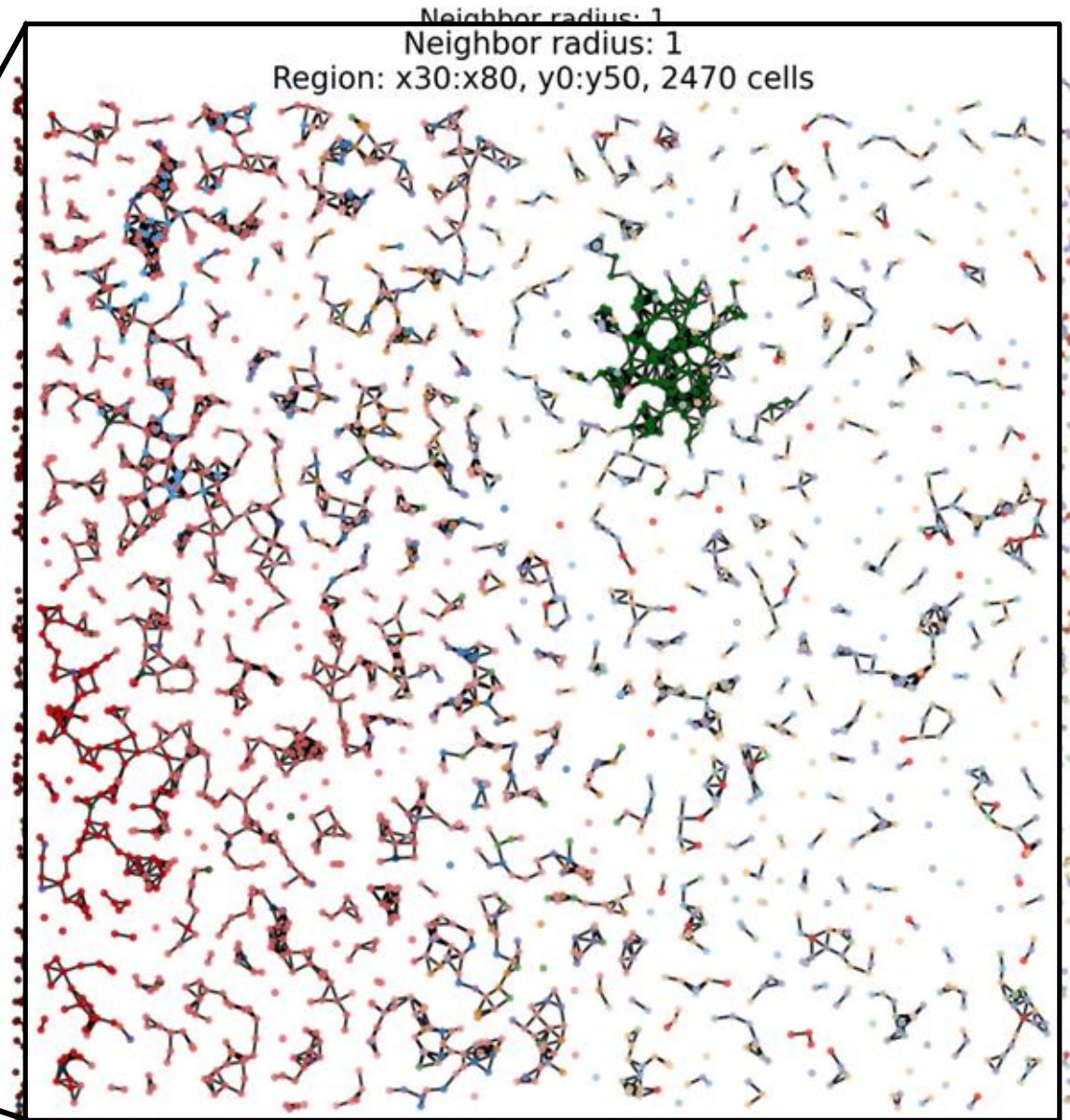
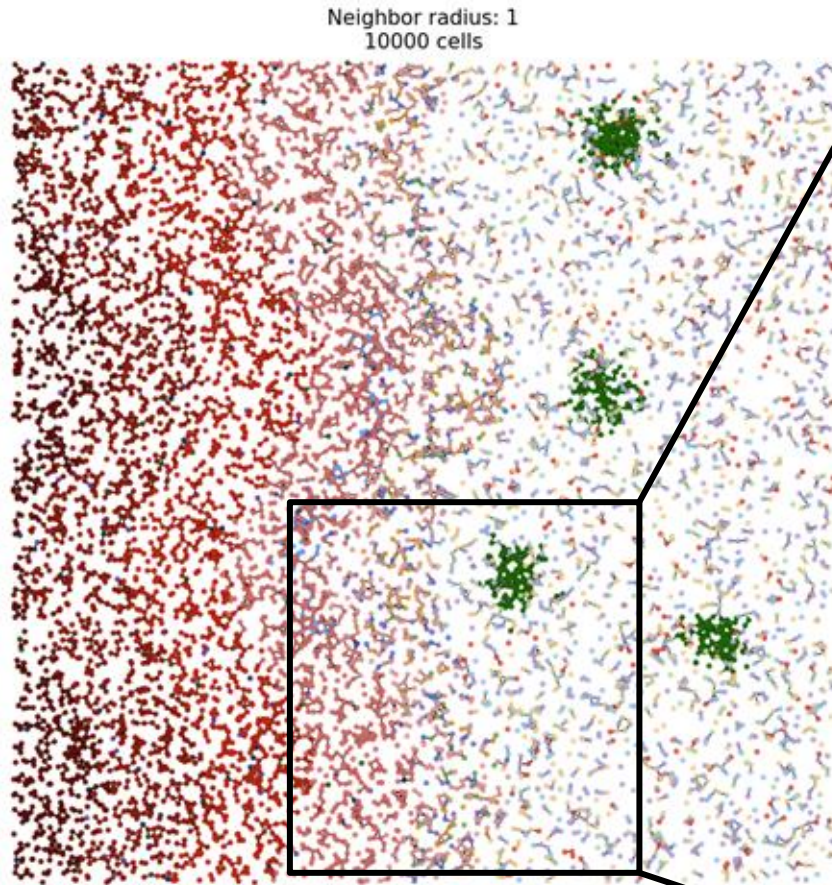
Xenium — segmented cells



A major consideration before benchmarking!

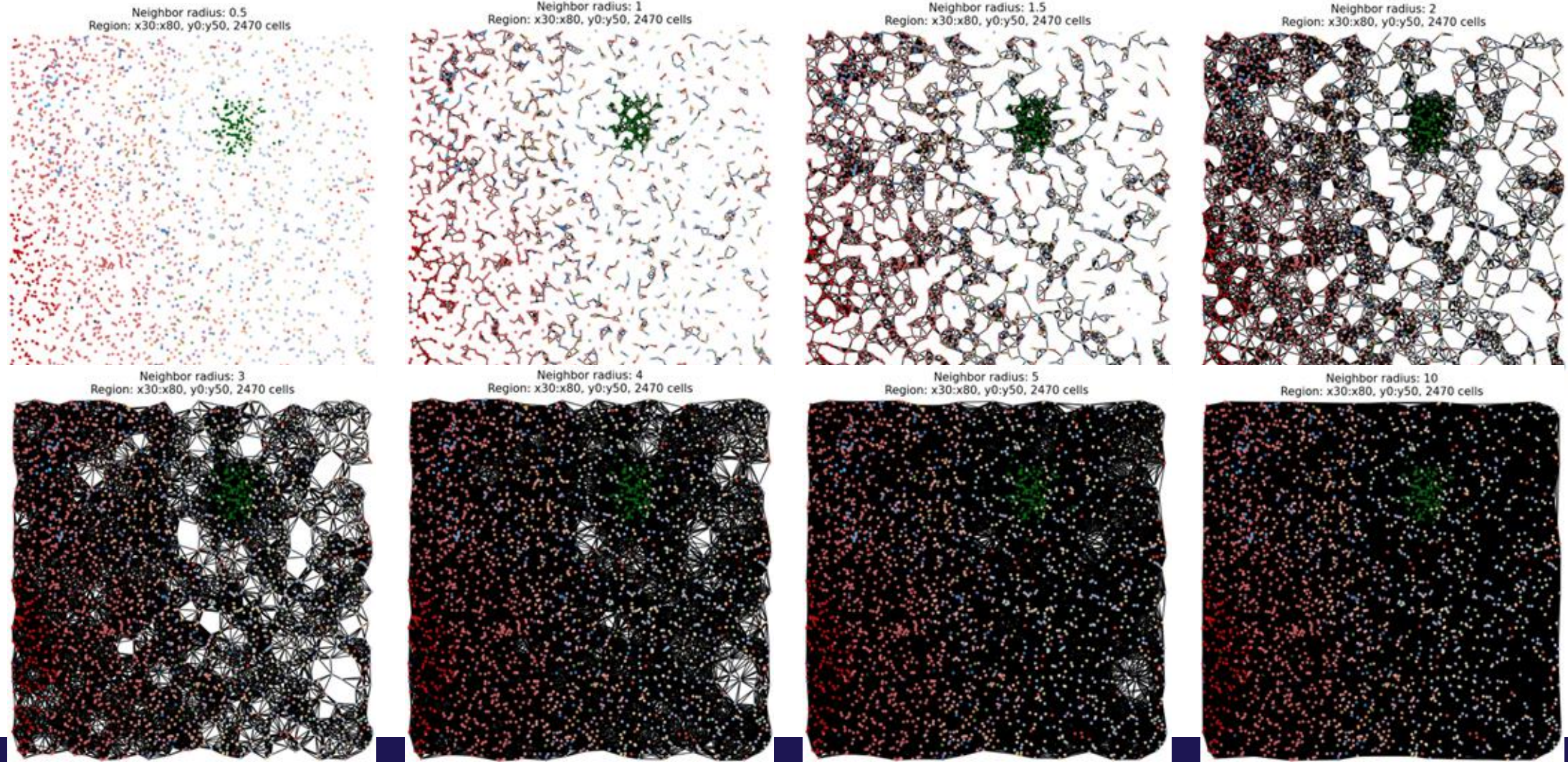
Exploring input graph data

Radius: 1, Coordinate scale: 1-100

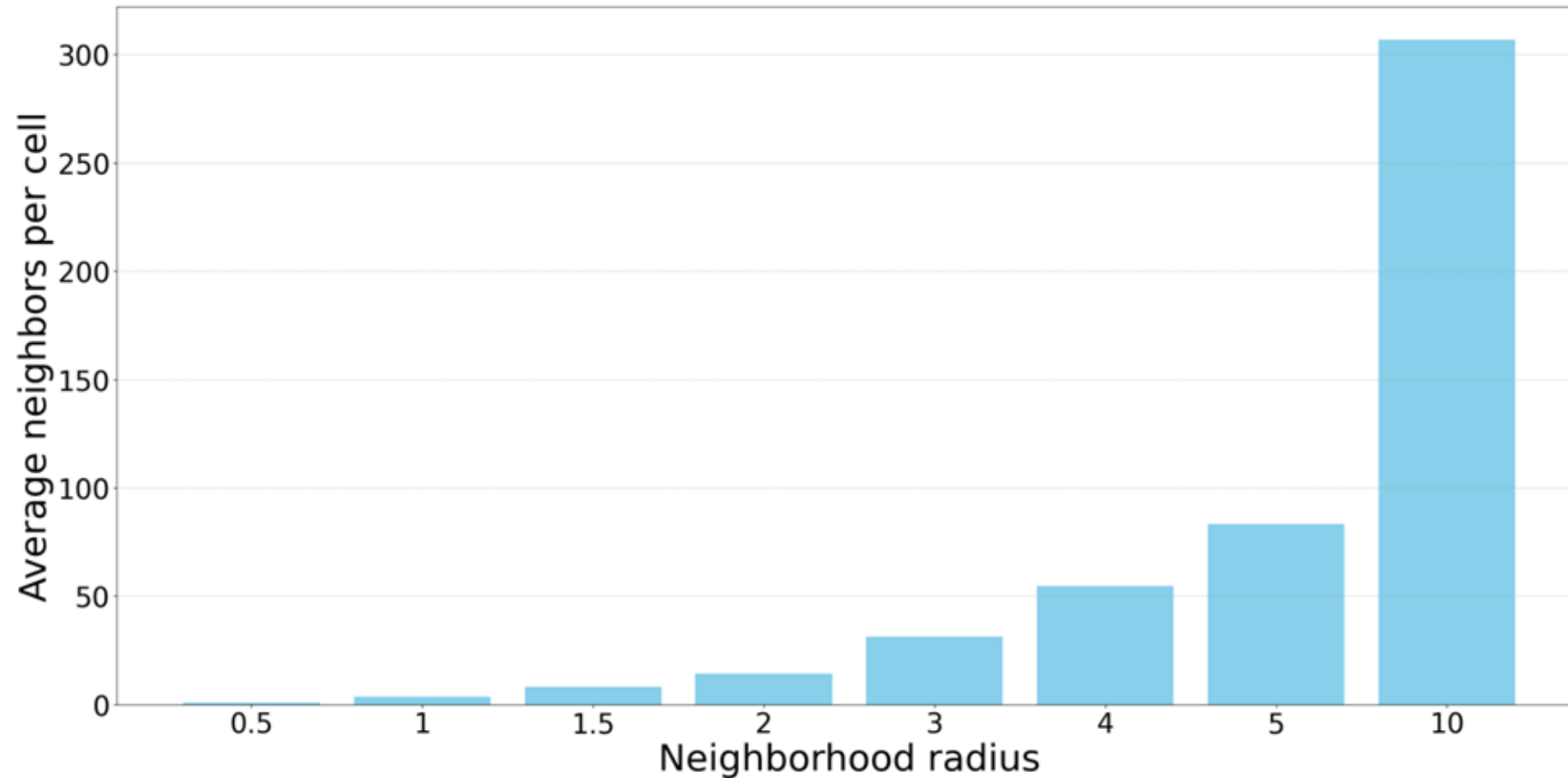


Selecting an appropriate neighborhood radius

Radius: 0.5, 1, 1.5, 2, 3, 4, 5, 10 (coordinate scale 1-100)



The graph can get “overconnected” very quickly!



Building the Graph with Squidpy

```
import squidpy as sq

# Build the spatial neighbours graph
sq.gr.spatial_neighbors(
    adata,
    coord_type='generic', # or 'grid'
    radius=10, # number most reasonable for data type
    delaunay=TRUE, # or FALSE
    n_neighs=6, # number of tiles
    spatial_key='spatial' # key where spatial coords stored
)

# Results stored in adata:
# adata.obsp['spatial_connectivities']
# adata.obsp['spatial_distances']
```

coord_type

'generic' for single-cell platforms (Xenium, MERFISH). Use 'grid' for spot-based (Visium).

radius

Neighbor distance cutoff.

delaunay

Whether to compute a graph from Delaunay triangulation.

The graph lives inside your AnnData object — ready for all squidpy analyses.

Graph-Based Metrics in Squidpy

Once the spatial graph is built, squidpy provides four key analysis functions:

`sq.gr.nhood_enrichment()`

Neighbourhood Enrichment

Are two cell types found next to each other more often than expected by chance?

`sq.gr.co_occurrence()`

Co-occurrence Score

How does the probability of finding a cell type change as a function of distance?

`sq.gr.centralty_scores()`

Centrality Scores

Which cell types are the most 'central' or well-connected in the graph?

`sq.gr.ripley()`

Ripley's Statistics

Is a cell type spatially clustered, dispersed, or randomly distributed?

Neighbourhood Enrichment

```
sq.gr.nhood_enrichment(adata, cluster_key='cell_type')
```

What it does:

Counts how often each pair of cell types are direct neighbours in the graph, then compares this count to what you would expect by chance (permutation testing).

Positive score = those cell types cluster together. Negative score = they avoid each other.

Why it matters:

Reveals immune exclusion zones, tumour-immune interfaces, and cellular niche organisation at single-cell resolution.

Output: Enrichment Heatmap

	Tumor	Immune	Stromal	Endo
Tumor	1.0	-0.8	0.2	-0.3
Immune	-0.8	1.0	0.6	0.1
Stromal	0.2	0.6	1.0	0.4
Endo	-0.3	0.1	0.4	1.0

Co-occurrence Score

```
sq.gr.co_occurrence(adata, cluster_key='cell_type')
```

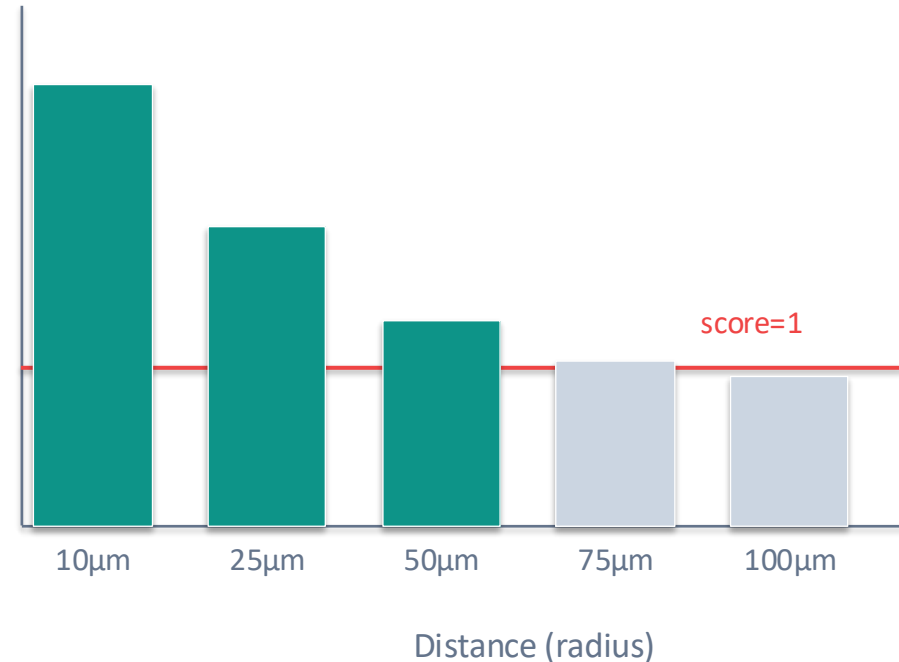
What it does:

Asks: "At each distance radius r , is cell type B more or less likely to be near cell type A than expected?"

Score > 1 = enrichment. Score $\sim$ 1 = random. Score < 1 = exclusion.

Unlike neighbourhood enrichment, this is continuous over distance — you can see how interactions strengthen or fade as you expand the radius.

Score vs. Distance (example)



Centrality Scores & Ripley's Statistics

`sq.gr.centraplity_scores()`

Measures how 'important' each cell type is within the graph topology.

Degree centrality

How many direct neighbours does a cell type have on average?

Betweenness centrality

How often does a cell type sit on the shortest path between two others?

Closeness centrality

How quickly can a cell type reach all others in the network?

Identifies hub cell types that mediate intercellular communication.

`sq.gr.ripley()`

Is a cell type spatially clustered, dispersed, or randomly distributed?

Ripley's L

Normalised version of Ripley's K — counts how many same-type cells fall within a radius r of each cell, across all radii

Ripley's G

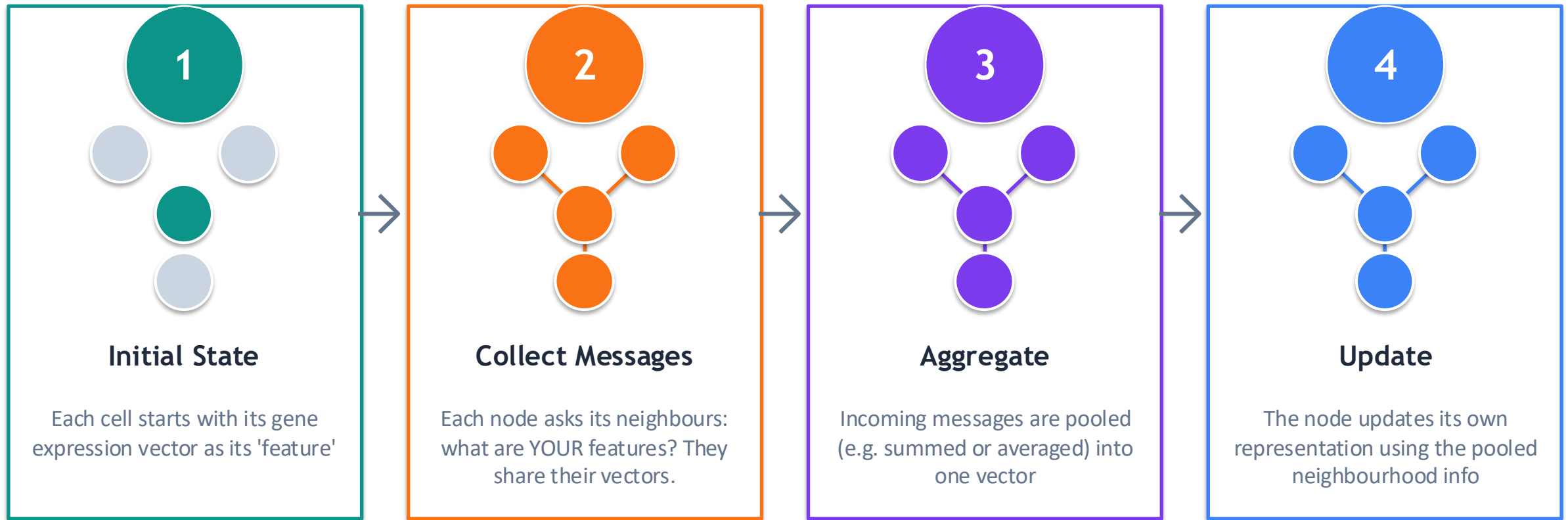
Distance to the nearest same-type neighbour — detects fine-scale clustering

Ripley's F

Distance from a random point to the nearest observed cell -- flips the question: instead of starting from a cell, it starts from a random point in space and asks how far you'd have to travel to hit any cell

How GNNs Work: Message Passing

Core idea: each node collects information from its neighbours, combines it, and updates its own representation.



Repeated across many layers: each node gradually accumulates context from wider and wider neighbourhoods

GNNs in Spatial Biology

What can we predict or discover using graph deep learning?

Spatial Domain Detection

Identify tissue regions (layers, niches) by learning from both gene expression AND neighbourhood composition at the same time.

Tools: STAGATE, GraphST

Cell Type Annotation

Propagate labels through the graph — a labelled cell teaches its unlabelled neighbours, exploiting spatial consistency.

Tools: scGNN, Seurat v5

Gene Expression Imputation

Predict expression of unmeasured genes using neighbouring cells as context — like inferring what a blurry pixel should look like from its surroundings.

Tools: SpaGE, gimVI

Cell-Cell Interaction Modelling

Learn which cell-type pairs communicate and predict downstream effects — going beyond co-occurrence to mechanistic inference.

Tools: NicheNet + GNN extensions

Previous work

Article | Published: 28 October 2021
SpaGCN: Integrating gene expression, spatial location and histology to identify spatial domains and spatially variable genes by graph convolutional network
Jian Hu¹, Xiangli Li, Kyle Coleman, Amelia Schroeder, Nan Ma, David J. Irwin, Edward B. Lee, Russell T. Shinohara & Miaoqiao Li¹
Nature Methods **18**, 1342–1351 (2021) | [Cite this article](#)

nature methods
Article
Unsupervised discovery of tissue architecture in multiplexed imaging
Received: 15 March 2022
Accepted: 21 September 2022
Published online: 12 October 2022
Junbum Kim¹, Samir Bustin¹, Juan Miguel Mosquera Benat Shaykhiev¹, André F. Rendeiro^{1,2,3,4} and Orit Rozenblatt-Kavan^{1,2,3,4}

Article | Open access | Published: 01 April 2022
Deciphering spatial domains from spatially transcriptomics with an adaptive graph attention encoder
Kangning Dong & Shihua Zhang¹
Nature Communications **13**, Article number: 1739 (2022) | [Cite this article](#)

Article | Open access | Published: 14 July 2022
Identifying multicellular spatiotemporal organization of cells with SpaceFlow
Honglei Ren, Benjamin L. Walker, Zixuan Cao & Qing Nie¹
Nature Communications **13**, Article number: 4076 (2022) | [Cite this article](#)

nature biomedical engineering
Article
Graph deep learning for the characterization of tumour microenvironments from spatial protein profiles in tissue specimens
Received: 16 May 2022
Accepted: 8 September 2022
Published online: 19 November 2022
Zhengqi Wu^{1,2}, Alessandro E. Trevino^{1,2,3}, Eric Wu^{1,2}, Kyle Swanson^{1,2}, Honesty J. Kim¹, H. Blaise D'Angelo¹, Ryan Preski¹, Gregory W. Charvillat¹, Piero D. Dalerba¹, Ann Marie Egloff¹, Ravindra Upgupta^{1,2}, Umamaheswar Duvvuri¹, Aaron T. Mayer^{1,2,3,4} & James Zou^{1,2,3,4}

nature genetics
Article
CellChart reveals spatial cell niches associated with tissue remodeling and cell plasticity
Received: 26 January 2023
Accepted: 23 October 2023
Published online: 08 December 2023
Marco Varrone^{1,2,3}, Daniele Tavernari^{1,2,3,4}, Albert Santamaria-Martinez^{1,2,3}, Logan A. Walsh^{1,2} & Giovanni Ciriello^{1,2,3,4}

nature methods
Article
Interpretable spatially aware dimension

Cell Reports Methods
Article
Discovery and generalization of tissue structures from spatial omics data
Graphical abstract
METHOD
Spatial Data Input Hybrid Graph Construction
Authors
Zhenqin Wu, Ayano Kondo, Monee McGrady, James Zou, Aaron T. Mayer, Alessandro E. Trevino
nature methods
Article
Interpretable spatially aware dimension

nature communications
Article
Identification and characterization of cell niches in tissue from spatial omics data at single-cell resolution
Received: 12 August 2024
Accepted: 3 February 2025
Published online: 16 February 2025
Jingyang Qian^{1,2,3}, Xin Shao^{1,2,3,4}, Hudong Bao^{1,2}, Yin Fang^{1,2}, Wenbo Guo^{1,2,3}, Chengyu Li^{1,2}, Anyao Li^{1,2}, Hua Hua^{1,2,3} & Xiaohui Fan^{1,2,3,4}

nature methods
Article
Tracking cells through time and space with cot
Received: 10 March 2024
Accepted: 11 May 2024
Published online: 22 January 2025
Domènec Klier^{1,2,3}, Giovanni Palla^{1,2,3}, Markus Lange^{1,2,3}, Mihai Klier^{1,2,3}, Zee Pines^{1,2}, Marcel Gander¹, Lantini Meng-Papayanash¹, Michael Beer¹, Lina Sabat^{1,2,3}, Changping Jiang^{1,2}, Anne Bastien-Poulin¹, Paula Costa^{1,2}, Maria Teresa Medeiros^{1,2}, Shrey Parikh¹, San Gelfi¹, Heiko Lickert^{1,2,3}, Mustafa Sahli^{1,2}, Moe Nisae^{1,2,3}, Marco Cusi^{1,2} & Fabian A. Theis^{1,2,3,4}

nature genetics
Article
Quantitative characterization of cell niches in spatially resolved omics data
Received: 15 March 2024
Accepted: 15 February 2025
Published online: 18 March 2025
updates
Sebastian Brk^{1,2,3,4}, Irene Bonafante-Pardoll^{1,2}, Adib Miraki Farid¹, Adam Boxall¹, Enderiz Agirre¹, Fani Memi¹, Anna Magusa^{1,2}, Anamika Yadav¹, Erick Armingol¹, Bong Fan^{1,2,3,4}, Gonçalo Castelo-Branco^{1,2}, Fabian J. Theis^{1,2,3,4}, Omer Ali Bayraktar¹, Carlos Talavera-López^{1,2,3} & Muhammad Lotfollahi^{1,2,3,4}

nature genetics
Article
Quantitative characterization of tissue niches using multiomics and ecological spatial analysis
Received: 24 June 2024
Accepted: 5 February 2025
Daisy Yi Ding^{1,2}, Zeyu Tang^{1,2,3}, Bokai Zhu^{1,2,3,4}, Hongyu Ren¹, Alex K. Shalek^{1,2,3,4}, Robert Tibshirani^{1,2,3} & Garry P. Nolan^{1,2,3,4}

nature methods
Article
Resolving tissue complexity by multimodal spatial omics modeling with MISO
Received: 20 January 2024
Accepted: 20 November 2024
Published online: 15 January 2025
Check for updates
Kyle Coleman^{1,2,3}, Amelia Schroeder¹, Melanie Loth¹, Daiwei Zhang¹, Jeong Hwan Park¹, Ji-Youn Sung¹, Niklas Blank^{1,2}, Alexis J. Cowan¹, Xiyu Qian¹, Jianfeng Chen¹, Rahul Jiang^{1,2}, Hanyang Yan¹, Laith Z. Samarrah¹, Jean R. Clementson¹, Inyong Jang¹, Mirgi Kim¹, Isabel Barnfather^{1,2}, Joshua D. Rabinowitz¹, Yanxiang Deng^{1,2,3}, Edward B. Lee^{1,2}, Alexander Lazar^{1,2}, Jianjun Gao¹, Emma E. Furth¹, Tae Hyun Hwang¹, Linghua Wang^{1,2,3}, Christoph A. Thuesen¹, Jian Hu^{1,2,3} & Mingyao Li^{1,2,3,4}

Key difference in graph-based methods

- Algorithmic
 - UTAG
 - Cellcharter
 - BANKSY
- Deep learning
 - STAGATE
 - GraphST

MENDER: fast and scalable tissue structure identification in spatial omics data

Received: 26 June 2023
Accepted: 11 December 2023
Zhiyuan Yuan^{1,2,3}

nature genetics
Article
BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis
Received: 3 April 2023
Accepted: 16 January 2024
Published online: 27 February 2024
Vipul Singhal^{1,2,3}, Nigel Chou^{1,2,3}, Joseph Lee^{1,2}, Yifei Yue¹, Jinyun Liu¹, Wan Kee Chook¹, Li Lin¹, Yun-Ching Chang¹, Erica Mei Ling Teo¹, Jonathan Awe¹, Hwee Kuan Lee^{1,2,3,4,5}, Kok Hao Chan^{1,2,3} & Shyam Prabhakar^{1,2,3,4}

SCIENCE ADVANCES | RESEARCH ARTICLE

CANCER

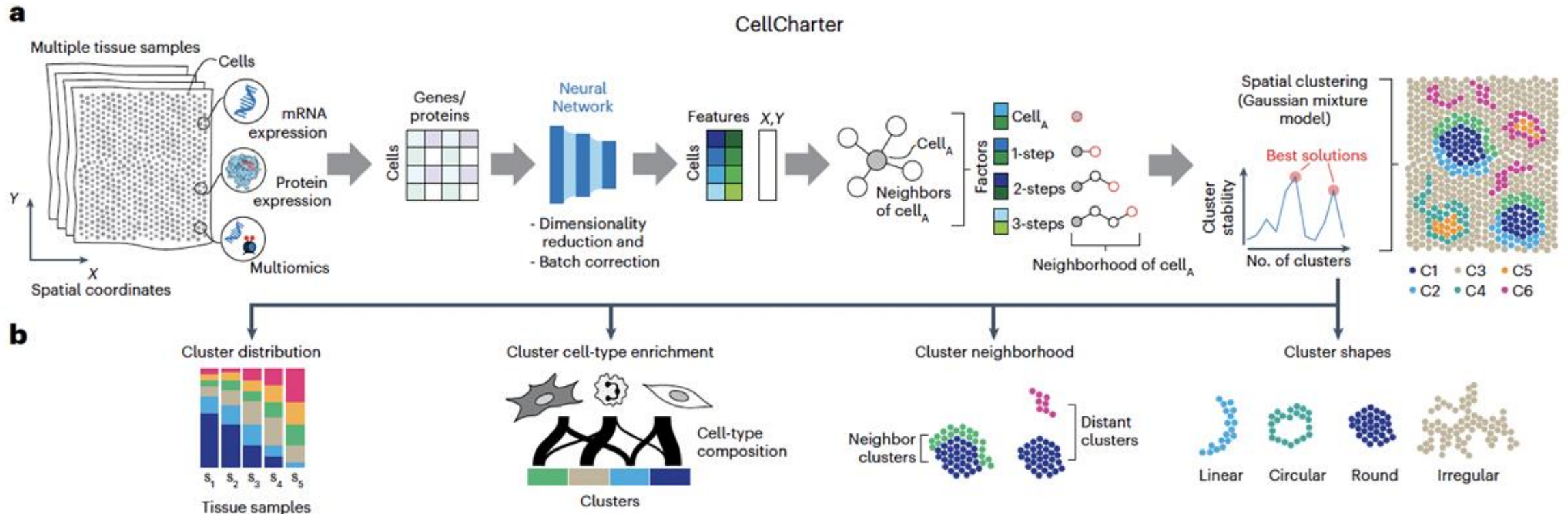
SpaTopic: A statistical learning framework for exploring tumor spatial architecture from spatially resolved transcriptomic data
Yueli Zhang^{1,2}, Bianjiong Yu^{1,2}, Wenxuan Ming^{1,2}, Xiaolong Zhou¹, Jin Wang¹, Dijun Chen^{1,2,3,4}

nature methods
Article
Mapping the topography of spatial gene expression with interpretable deep learning
Received: 9 October 2023
Accepted: 14 October 2024
Uthavi Chitra¹, Brian J. Arnold¹, Hirak Sarkar^{1,2}, Kohei Sanno¹, Cong Ma¹, Sereño Lopez-Darvel¹ & Benjamin J. Raphael^{1,2,3}

CellCharter

Variational autoencoder

- Dimensionality reduction recommendation differs based on data type
- Neighborhood layers configurable
- Many tools “pieced” together (eg., scvi, squidpy, gmm)



STAGATE

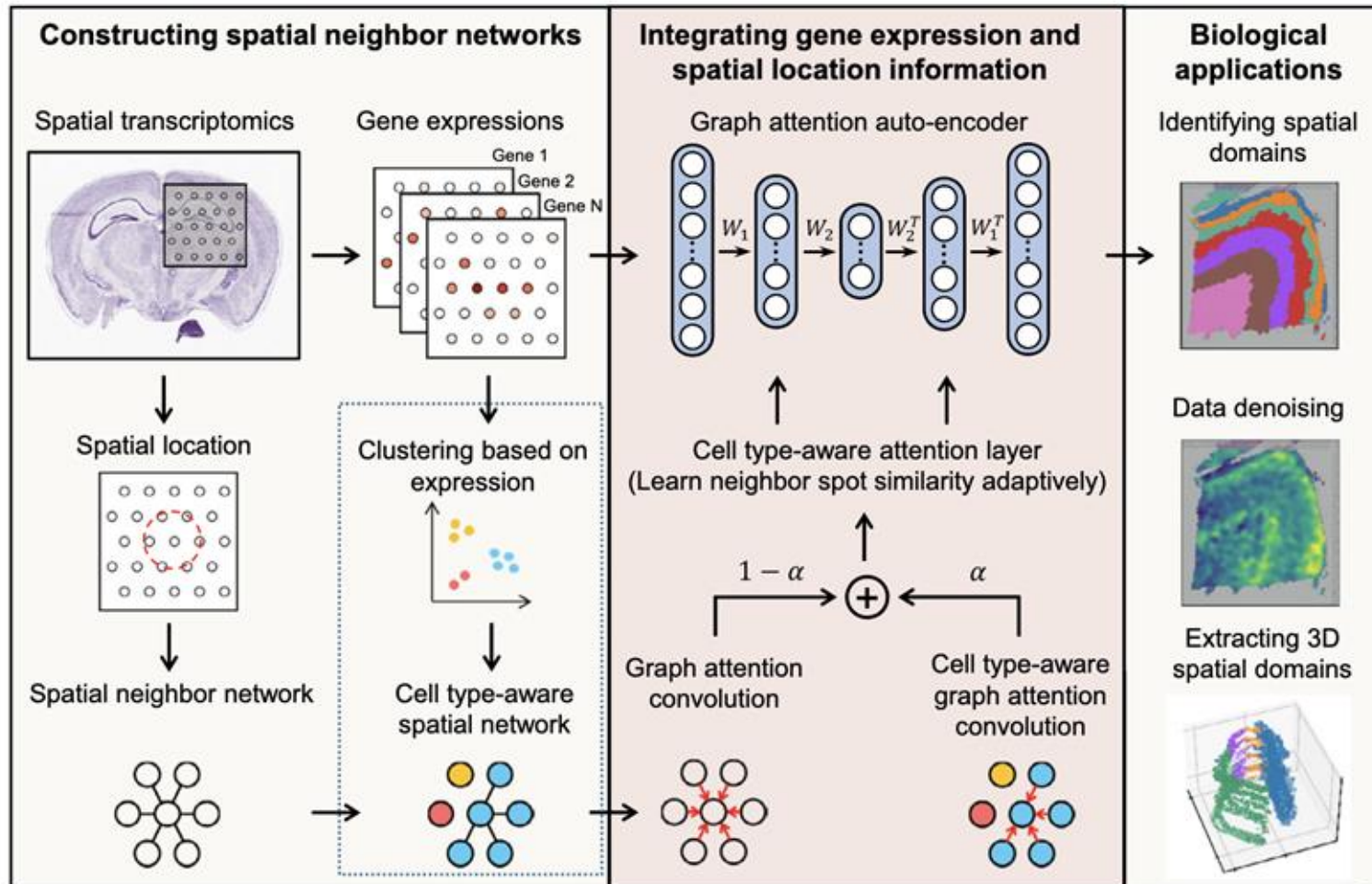
Graph Attention Autoencoder

Article [Open access](#) | Published: 01 April 2022

Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder

Kangning Dong & Shihua Zhang 

Nature Communications 13, Article number: 1739 (2022) | [Cite this article](#)



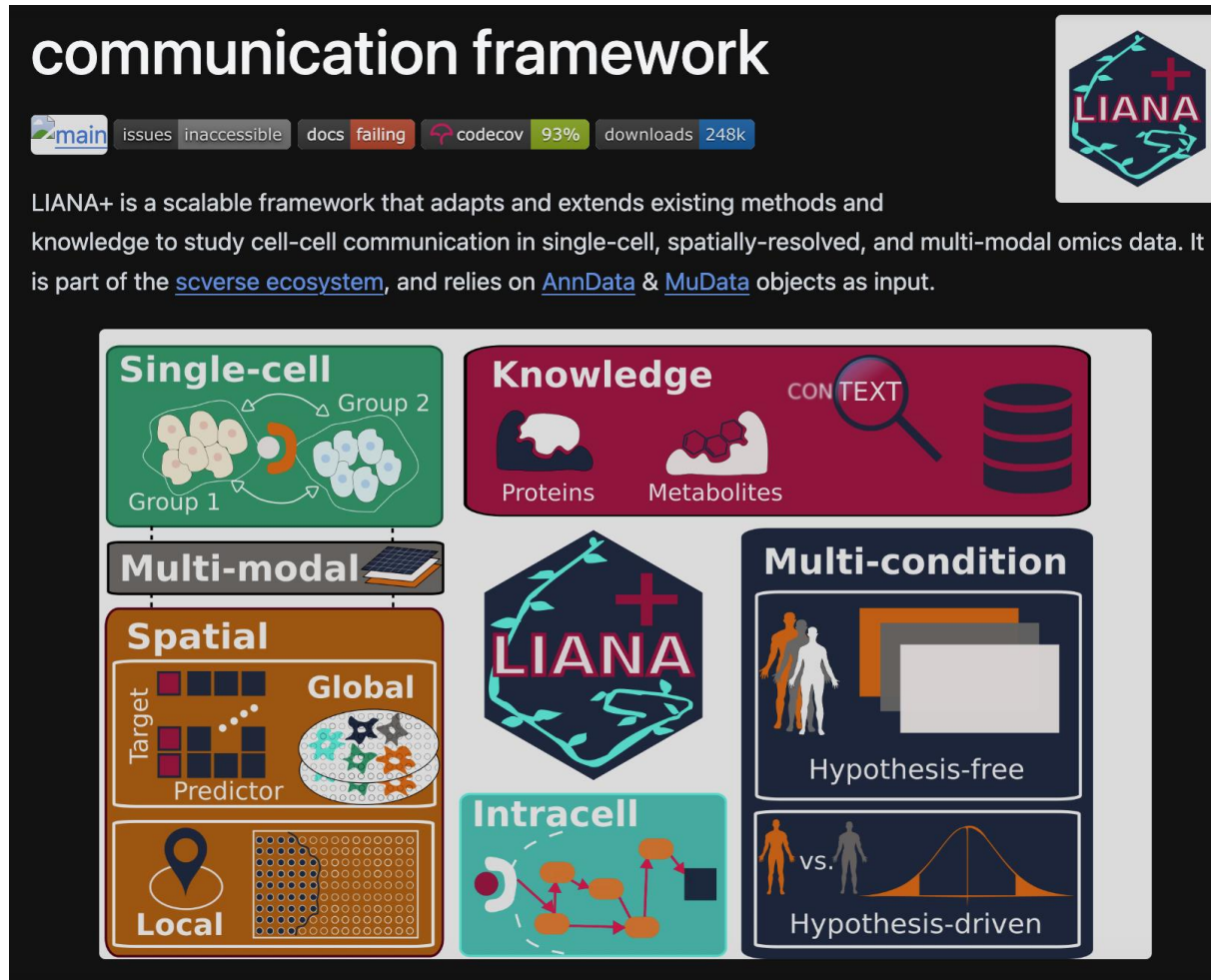
- Encoder uses message passing, decoder reconstructs expression
- Constructs a spatial neighborhood graph and optionally refines it with a cell type-aware module

CellCharter (algorithmic) vs STAGATE (deep learning)

- **STAGATE** is a true graph neural network
 - Spatial graph is baked into the model architecture itself and uses graph attention
 - During training, the model *learns* how much to weight each neighbor
 - The neighborhood aggregation and the representation learning happen simultaneously
- **CellCharter** is not a GNN but a standard variational autoencoder (built on scVI)
 - Spatial context is injected as a preprocessing step
 - Neighboring cells' expression is averaged over a few “hops” and concatenated into each cell's “profile”
 - The aggregation is fixed and unweighted

STAGATE learns how to aggregate neighbors;
CellCharter decides upfront and then forgets about the graph

Beyond motifs: Cell-cell communication



Instead of just asking "do cell type A and cell type B express a ligand-receptor pair?" it weights those interactions by **how physically close** the two cell types actually are in tissue — using the spatial graph.

So a ligand-receptor pair only scores highly if the cells expressing it are also spatially proximal.

Key Takeaways

1

Graphs encode spatial context — cells as nodes, proximity as edges

2

Squidpy builds and analyses the graph automatically from your AnnData

3

Graph metrics reveal neighbourhoods, co-occurrence, clustering, and centrality

4

GNNs extend deep learning to graphs, learning from expression AND spatial structure
